

FRIDAY, AUGUST 28TH, 2026

CROSS-DOMAIN SYNTHESIS

The Threshold at One — Criticality in Six Sciences

Six fields arrived independently at the same dividing line between dying out and running away, and every one of them wrote it down as the same single line of arithmetic.

3,084 words · 16-minute read

In 1873 Francis Galton put a question to the readers of the *Educational Times*. He had noticed that the surnames of eminent families kept disappearing, and wanted to know how much of that was decline and how much was luck. Suppose each man has some number of sons, drawn independently from a fixed distribution. What is the probability that his male line has died out after a given number of generations?

The question was Victorian in its preoccupations and entirely modern in its structure: a population whose members reproduce independently, whose fate therefore depends not on any individual but on an average. A clergyman and mathematician named Henry William Watson replied, and in 1875 the two published the answer jointly in the *Journal of the Anthropological Institute* [1]. Their formulation — one ancestor, independent offspring, a distribution that does not change from generation to generation — is now called the Galton-Watson process, and it is the skeleton of a remarkable number of things.

The paper also got the answer wrong. Watson concluded that extinction is certain, always, for every offspring distribution — including the ones describing populations that in fact grow without bound. That is not a small slip; it is the whole result, inverted.

An Argument About Surnames

Watson's mistake was an error about a fixed point, and the correct treatment took fifty-five years to appear in print. In 1927 J. B. S. Haldane, working on the survival of new mutations rather than of surnames, gave a much more careful account and derived a result that reappears later in this piece [2]. In 1929 the Danish telephone engineer A. K. Erlang, who had reinvented the problem while thinking about something else, posed it again; his compatriot Johan Frederik Steffensen answered it properly in 1930, in Danish, in *Matematisk Tidsskrift* [3]. Writing in Danish is an effective way to be ignored, and Steffensen reissued the work in French three years later [4]. The result is sometimes called the Galton-Watson-Haldane-Steffensen criticality theorem, which is an unusually honest piece of naming.

Even that history has a floor under it. In 1845, twenty-eight years before Galton asked his question, the French statistician Irénée-Jules Bienaymé published a note on the law of multiplication and the duration of families that stated the correct criterion outright. He promised the derivation in a later paper, and so

far as anyone has established, never published it. The priority went unnoticed until Heyde and Seneta reconstructed it in 1972 [5]. The mathematics of extinction, in other words, was first stated correctly by someone who did not show his work, then named after someone who showed his work and got it wrong.

The Fixed Point

The modern machinery is compact enough to state in full. Let p_j be the probability that an individual has exactly j offspring, and collect those probabilities into a probability generating function,

$$f(s) = \sum_{j \geq 0} p_j s^j, \quad 0 \leq s \leq 1.$$

Two facts make this the right object. The generating function of a sum of independent copies is the product of their generating functions, so composing f with itself advances the process one generation; and $f'(1)$ is the mean number of offspring, written m . Let q be the probability that a line starting from one ancestor eventually dies out. That ancestor has some number of children, and the line dies out precisely when all of their lines die out independently, which gives the self-consistency condition

$$q = f(q).$$

The extinction probability is the *smallest* root of that equation in the unit interval $[3]$, $[4]$. Since f is increasing and convex with $f(1) = 1$, the geometry is forced. If the slope at $s = 1$ exceeds one, the curve must cross the diagonal exactly once before reaching the corner, and that crossing is the extinction probability; if the slope is at most one, the curve stays above the diagonal and the only root is $q = 1$.

That is the entire theorem. Extinction is certain if and only if $m \leq 1$. Above one, survival has positive probability; below one, no matter how patient anyone is, the line ends. The threshold sits at the mean equal to one and depends on nothing else about the distribution.

When offspring are Poisson with mean m — the natural law when each individual has many independent, low-probability chances to reproduce — the condition becomes a single transcendental equation,

$$q = e^{-m(1-q)},$$

which has no closed form but converges in a few iterations. At $m = 1.5$ it gives $q = 0.4172$, so a single ancestor has a 58.3 per cent chance of founding a line that never ends. At $m = 3$ it gives $q = 0.0595$.

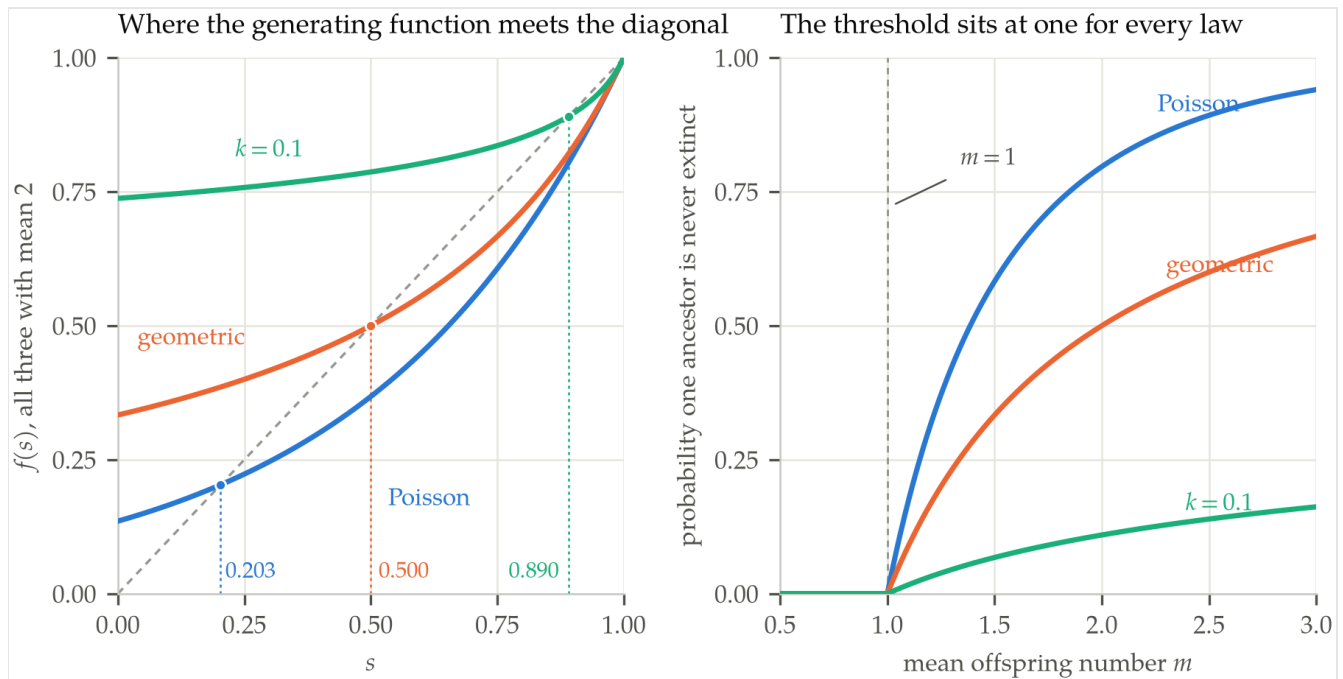


FIGURE The extinction probability is the first crossing of the generating function with the diagonal. All three laws here have mean 2, so all three share the same tangent at the corner and the same threshold, and they disagree about the outcome by a factor of four.

The Same Condition, Six Vocabularies

What makes this worth an hour is not the theorem, which is a page of undergraduate analysis, but that six sciences walked into it separately, in different decades, without recognising each other. Two are already in hand: demography, and through Haldane population genetics. Here are the other four.

Consider polymer chemistry. Paul Flory, asking why liquid resins abruptly become solid gels at a well-defined moment in a reaction, reasoned in 1941 that a branching unit of functionality f , having already been reached by one chain, can send out $f - 1$ further chains, each present with probability α ; an infinite network becomes possible exactly when $\alpha(f - 1) > 1$ [6]. That is the branching condition verbatim, with $\alpha(f - 1)$ as m . The critical conversion is $\alpha_c = 1/(f - 1)$ — one half for a trifunctional monomer, one third for a tetrafunctional one, the values Flory reported and Stockmayer extended two years later [7]. Nothing in the derivation is about chemistry.

Consider nuclear engineering. On 4 July 1934, five years before fission was understood at all, Leo Szilard filed a British patent application describing a neutron chain reaction and, with it, the concept of a critical mass: a quantity of material below which each neutron generation is smaller than the last, and above which it is not. He asked the Patent Office to withhold publication, assigned the patent to the British Admiralty, and GB 630,726 stayed secret for thirteen years [8]. The reactor physicist's m is the effective multiplication factor k_{eff} , fission-causing neutrons produced per fission-causing neutron consumed, and the critical condition is $k_{\text{eff}} = 1$.

Ten years later, and this time deliberately, the general theory was written down at Los Alamos. David Hawkins and Stanisław Ulam circulated *Theory of Multiplicative Processes* as report LA-171 on 14 November 1944, declassified in 1946; Ulam and C. J. Everett extended it in three further reports in 1948. The motivation was neutron multiplication in a bomb; the framing was not. The editors of Ulam’s collected Los Alamos reports describe the work as laying foundations for a formalism general enough to include, as special cases, the multiplication of bacteria, radioactive decay, cosmic ray showers, diffusion theory and the theory of trajectories in mechanical systems [9]. The generality was visible from inside the problem, to people who had never heard of Galton or Watson.

Consider epidemiology. Kermack and McKendrick’s 1927 paper derived a threshold population density below which no epidemic can occur, from a deterministic model with no branching process in it anywhere [10]. Whittle supplied the stochastic version in 1955, sandwiching the early phase of an epidemic between two birth-and-death processes and showing that the probability an introduction “takes” is exactly a branching-process survival probability [11]. The epidemiological name for m is the basic reproduction number R_0 , and the entire logic of vaccination follows from the threshold. Immunising a fraction v of a population multiplies the effective reproduction number by $1 - v$, so the coverage that pushes the mean below one is $v^* = 1 - 1/R_0$.

Consider network science. Erdős and Rényi showed in 1960 that a random graph on n vertices with mean degree c contains a component of size proportional to n if and only if $c > 1$ [12]. The fraction S of vertices in that giant component satisfies $S = 1 - e^{-cS}$, the Poisson extinction equation with the sign of the unknown flipped, and it returns the same number: at $c = 1.5$ the giant component holds 58.3 per cent of the graph, exactly the survival probability at $m = 1.5$. It is the same equation because it is the same process — following a network outward from a random vertex is a branching process whose offspring law is the distribution of *further* edges at each step. Made precise by Molloy and Reed in 1995 and by Newman, Strogatz and Watts in 2001, the criterion for a general degree distribution is $\sum_k k(k-2)p_k > 0$ [13], [14], or $\langle k^2 \rangle / \langle k \rangle - 1 > 1$. The left-hand side is the mean of the excess-degree distribution: new edges reached per edge traversed. It is m , again.

What the Mean Does Not Tell You

Having established that one number decides everything, it is worth being precise about how little it decides. The threshold is a property of the mean alone. Nothing else is.

Expand the fixed-point equation just above the threshold, writing $m = 1 + \varepsilon$ and letting σ^2 be the offspring variance. The survival probability behaves as

$$S \approx \frac{2(m-1)}{\sigma^2},$$

which is the sharpest one-line statement of the situation: the threshold’s *location* is set by the mean, the *slope* with which survival lifts off it by the variance. For Poisson offspring the variance equals the mean, so $S \approx 2\varepsilon$, and a mutation conferring a one per cent fitness advantage survives with probability about

two per cent. That is Haldane’s 1927 result; the exact fixed point at $m = 1.01$ gives 0.0197 against the approximation’s 0.0198.

Away from the threshold the divergence becomes enormous. Fix the mean at exactly two and vary only the shape.

Offspring law	Variance	Extinction from one ancestor
One, two or three, equally likely	0.67	0.000
Poisson	2.0	0.203
Geometric	6.0	0.500
Negative binomial, $k = 0.1$	42.0	0.890

Every row has the same threshold and the same mean. The first cannot go extinct at all, because nobody is ever childless. The last goes extinct nine times in ten, because most individuals have no offspring whatever and the average of two is carried by a rare few with dozens.

WHY THE SAME AVERAGE BEHAVES SO DIFFERENTLY

Two epidemics have $R_0 = 2$. In the first, every infected person infects one, two or three others. In the second, ninety per cent infect nobody and the rest infect twenty. Both have the same threshold, the same long-run growth rate once established, and the same herd-immunity requirement. What differs?

What differs is the fate of a single introduction. The first epidemic, once seeded, always establishes; the second dies out about nine times in ten. Nor is this hypothetical. Lloyd-Smith and colleagues assembled contact-tracing data for eight directly transmitted diseases and found individual infectiousness strongly skewed, fitting a negative binomial with dispersion parameter $k = 0.16$ for the 2003 SARS outbreak in Singapore [15]. Endo and colleagues estimated $k \approx 0.1$ for SARS-CoV-2 from cluster sizes outside China, with a 95 per cent credible interval of 0.04 to 0.2 — roughly ten per cent of cases producing eighty per cent of infections [16]. At $R_0 = 3$ and $k = 0.1$ the variance-to-mean ratio is thirty-one, and a single imported case ignites a sustained outbreak only sixteen per cent of the time.

Most introductions of a superspreading pathogen therefore fizzle, which buys time; the ones that do not begin with a large first burst rather than a small one. Quiet months punctuated by an explosion are not evidence that anything changed. They are what a single sample from a heavy-tailed offspring law looks like.

One notational warning, because the fields collided. In epidemiology k is the dispersion parameter and small k means high variance; in reactor physics k is the multiplication factor and it is the mean. They are unrelated.

Twenty-Eight Minutes Under Stagg Field

On the afternoon of 2 December 1942, in a squash court under the stands of Stagg Field at the University of Chicago, Enrico Fermi's team withdrew the last control rod from a pile of graphite and uranium. The reaction ran for twenty-eight minutes at a doubling time of about two minutes, rising by a factor of roughly sixteen thousand, and reached about half a watt before the rod went back in [17], [18].

Those figures check each other: twenty-eight minutes at two minutes per doubling is fourteen doublings, and $2^{14} = 16,384$. Given that, the multiplication factor can be recovered rather than taken on trust. The asymptotic period is the doubling time divided by $\ln 2$, or 173 seconds, and in the one-group approximation

$$T = \frac{\ell}{k_{\text{eff}} - 1},$$

where ℓ is the mean time between neutron generations, and this is where reactor physics becomes an engineering discipline rather than a branch of demolition. Most fission neutrons are prompt, appearing within about 10^{-4} seconds; a fraction $\beta \approx 0.0065$ in uranium-235 arrive delayed, emitted by fission fragments decaying over seconds, with a mean precursor lifetime near 12.5 seconds. Averaging gives $\ell \approx 10^{-4}(1 - \beta) + \beta/\bar{\lambda} \approx 0.081$ seconds — eight hundred times the prompt lifetime, because 0.65 per cent of the neutrons take a hundred thousand times as long to arrive. So $k_{\text{eff}} - 1 = 0.081/173 = 4.7 \times 10^{-4}$: Fermi's pile was running five parts in ten thousand above the threshold.

Had those neutrons all been prompt, the same k_{eff} would have given a doubling time of 0.15 seconds and the pile's fourteen doublings would have been over in two seconds. The reactor therefore has a second threshold above the first: at $k_{\text{eff}} = 1 + \beta = 1.0065$ the prompt neutrons alone are self-sustaining, the delayed ones stop mattering, and control by moving a rod becomes impossible. Everything an operator does lives in the sliver of width 0.0065 between the two.

The deeper contrast with an epidemic is not the offspring law but the sample size. At half a watt, with about 200 MeV released per fission, the pile was running

$$\frac{0.5 \text{ W}}{200 \text{ MeV}} = 1.6 \times 10^{10} \text{ fissions per second,}$$

which is to say sixteen billion simultaneous branching processes a second, all with the same mean. The law of large numbers annihilates the variance, the observable behaves deterministically, and a slide rule is an adequate instrument. An epidemiologist watching a single imported case has a sample size of one. Same theorem, same threshold, incomparable epistemic situations — which is why criticality in a reactor is a number you read off a dial and criticality in an outbreak is something argued about for weeks afterwards.

The reactor's determinism is not the same as safety. On 30 September 1999 at the JCO fuel plant in Tokai-mura, workers pouring an 18.8 per cent enriched uranyl nitrate solution into a precipitation vessel by hand exceeded the critical mass. The solution went critical, boiled and expanded, dropped subcritical, cooled and went critical again, oscillating intermittently for about twenty hours. Three operators were overexposed and two died [19].

The Threshold Is Also an Attractor

Tokai-mura points at the last structural fact, and the most easily missed. In systems that consume the resource they grow on, the threshold is not a line the system happens to sit on one side of. It is where the system ends up.

Kermack and McKendrick saw this in 1927 and stated it with some relish: if the population density slightly exceeds its threshold value, the effect of an epidemic will be to reduce the density as far below the threshold as it initially was above it [10]. An epidemic burns through susceptibles, dragging the effective reproduction number down through one — it peaks exactly when $R_{\text{eff}} = 1$, and then overshoots. A reactor with negative temperature feedback does the same on a timescale of seconds, which is what produced the oscillation at Tokai-mura. A gelling resin consumes the reactive groups that make gelation possible. Criticality is a repeated destination, not a coincidence.

Which is why the threshold is a policy object. Measles is about as transmissible as any human pathogen that has been carefully measured; a systematic review of the literature puts its basic reproduction number at 12 to 18 [20], giving a herd-immunity threshold of 91.7 to 94.4 per cent — high enough that no vaccination programme has much slack. In 2024, WHO and UNICEF estimated global coverage at 84 per cent for the first measles dose and 76 per cent for the second, against a 90 per cent target, and at 66 and 52 per cent respectively in low-income countries [21]. Taking $R_0 = 15$, a population with 76 per cent immunity has an effective reproduction number of 3.6. The threshold does not negotiate, and the arithmetic of falling one dose short is not gradual.

There is one genuine escape from the framework, and it comes from the network side. The branching mean on a network is the excess-degree average $\langle k^2 \rangle / \langle k \rangle - 1$, which depends on the *second* moment of the degree distribution. For a power-law degree distribution with a sufficiently heavy tail, that second moment diverges as the network grows, and the epidemic threshold goes to zero: any transmissibility at all, however small, is enough. Pastor-Satorras and Vespignani established this in 2001, and it remains the sharpest statement of why highly heterogeneous contact structures behave qualitatively differently from homogeneous ones [22]. It is not an exception to the theorem. It is the theorem, applied to a distribution whose variance is infinite — a reminder that the quantity doing the work was never really the mean by itself.

The relationship between a surname, a resin, a neutron and a virus is not an analogy in the loose sense, of unrelated things said to be a bit alike. It is an identity of structure: independent replication, one number summarising it, a sharp transition where that number crosses one. Six fields found it separately

because it was there to be found, and the one that got it named got it backwards.

REFERENCES

- [1] H. W. Watson and F. Galton, "On the Probability of the Extinction of Families," *Journal of the Anthropological Institute*, vol. 4, pp. 138-144, 1875.
- [2] J. B. S. Haldane, "A Mathematical Theory of Natural and Artificial Selection, Part V," *Proceedings of the Cambridge Philosophical Society*, vol. 23, pp. 838-844, 1927.
- [3] J. F. Steffensen, "Om Sandsynligheden for at Afkommet uddør," *Matematisk Tidsskrift B*, pp. 19-23, 1930.
- [4] J. F. Steffensen, "Deux problèmes du calcul des probabilités," *Annales de l'Institut Henri Poincaré*, vol. 3, pp. 319-344, 1933.
- [5] C. C. Heyde and E. Seneta, "The simple branching process, a turning point test and a fundamental inequality: A historical note on I. J. Bienaymé," *Biometrika*, vol. 59, no. 3, pp. 680-683, 1972.
- [6] P. J. Flory, "Molecular Size Distribution in Three Dimensional Polymers. I. Gelation," *Journal of the American Chemical Society*, vol. 63, pp. 3083-3090, 1941.
- [7] W. H. Stockmayer, "Theory of Molecular Size Distribution and Gel Formation in Branched-Chain Polymers," *Journal of Chemical Physics*, vol. 11, pp. 45-55, 1943.
- [8] L. Szilard, "Improvements in or relating to the Transmutation of Chemical Elements," British Patent GB 630,726, filed 4 July 1934, published 1949.
- [9] A. R. Bednarek and F. Ulam, Eds., *Analogies Between Analogies: The Mathematical Reports of S. M. Ulam and his Los Alamos Collaborators*. Berkeley: Univ. of California Press, 1990.
- [10] W. O. Kermack and A. G. McKendrick, "A Contribution to the Mathematical Theory of Epidemics," *Proceedings of the Royal Society A*, vol. 115, pp. 700-721, 1927.
- [11] P. Whittle, "The Outcome of a Stochastic Epidemic — A Note on Bailey's Paper," *Biometrika*, vol. 42, no. 1-2, pp. 116-122, 1955.
- [12] P. Erdős and A. Rényi, "On the Evolution of Random Graphs," *Publications of the Mathematical Institute of the Hungarian Academy of Sciences*, vol. 5, pp. 17-61, 1960.
- [13] M. Molloy and B. Reed, "A Critical Point for Random Graphs with a Given Degree Sequence," *Random Structures and Algorithms*, vol. 6, no. 2-3, pp. 161-180, 1995.
- [14] M. E. J. Newman, S. H. Strogatz and D. J. Watts, "Random Graphs with Arbitrary Degree Distributions and Their Applications," *Physical Review E*, vol. 64, art. 026118, 2001.
- [15] J. O. Lloyd-Smith, S. J. Schreiber, P. E. Kopp and W. M. Getz, "Superspreading and the Effect of Individual Variation on Disease Emergence," *Nature*, vol. 438, pp. 355-359, 2005.
- [16] A. Endo, S. Abbott, A. J. Kucharski and S. Funk, "Estimating the Overdispersion in COVID-19 Transmission Using Outbreak Sizes Outside China," *Wellcome Open Research*, vol. 5, art. 67, 2020.
- [17] C. Allardice and E. R. Trapnell, "The First Pile," U.S. Atomic Energy Commission; reprinted in *IAEA Bulletin*, vol. 4, no. 5, 1962.
- [18] Argonne National Laboratory, "Chicago Pile 1: A Bold Nuclear Physics Experiment with Enduring Impact," 2017.
- [19] International Atomic Energy Agency, "Lessons Learned from the JCO Nuclear Criticality Accident in Japan in 1999," Vienna, 2000.
- [20] F. M. Guerra et al., "The Basic Reproduction Number (R_0) of Measles: A Systematic Review," *The Lancet Infectious Diseases*, vol. 17, no. 12, pp. e420-e428, 2017.
- [21] World Health Organization and UNICEF, "Immunization Coverage," fact sheet, 2024 estimates, 15 July 2025.
- [22] R. Pastor-Satorras and A. Vespignani, "Epidemic Spreading in Scale-Free Networks," *Physical Review Letters*, vol. 86, no. 14, pp. 3200-3203, 2001.

Daily Learning No. 014 · Cross-Domain Synthesis · Friday, August 28th, 2026 · Sources are listed above; every load-bearing figure traces to a numbered reference.